

Lab 02 - First Verilog

In this lab, you've learned how to do an initial and simple design in Verilog to learn the Vivado tooling and process involved in RTL/FPGA design.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

We learned how to setup a project in Vivado, plug into a circuit board, and create code which interacts with the physical board.

Lab Questions

1 - Describe the stages of building a Verilog project in Vivado.

We first wrote the verilog top and test files, and imported them as well as the constraint files. Once the files were in and the board was connected, we ran the simulation. After the simulation showed what we wanted, we ran the synthesis, implementation, and generated the bitstream. Once all of that was completed, we were able to run the verilog files to interact with the physical board.

2 - What is the value in looking at the elaborated design schematic?

It's valuable to look at the design schematic to see how what you're doing maps onto the physical board before running everything. It will tell you valuable information without having to take a bunch of time running the synthesis and implementation for every little change. Very obvious errors will be caught in the schematic and save a ton of time.

3 - Why should we simulate our designs frequently? What does the simulation do?

We should simulate our designs frequently to make sure your code does what you expect it to do, and to update any failures. It would take a very long time to program if each time you wanted to test your code you had to program the device. Simulating it is a much faster process.

Code Submission

Upload a .zip of all your code or a public repository on GitHub.

<https://github.com/CanonMiller/ece230>